

Hands-On Lab: Amazon Inspector (EC2 + ECR)

 **Goal:** Automatically detect vulnerabilities in EC2 and container images

Step-by-Step (for EC2)

Step 1: Enable Inspector

1. Go to **Amazon Inspector** in AWS Console
2. Click **Enable Inspector**

Step 2: Ensure EC2 is Managed by Systems Manager

- EC2 must have:
 - IAM role with `AmazonSSMManagedInstanceCore`
 - SSM Agent installed and running
 - Connectivity to SSM endpoints

Step 3: Wait for Scan Results

- Inspector will automatically scan and list findings:
 - Critical vulnerabilities (e.g., CVE-2023-XXXXX)
 - OS packages needing patches

Step-by-Step (for ECR)

Step 1: Push a container image to Amazon ECR

```
docker build -t test-image .  
docker tag test-image <your-ecr-uri>  
docker push <your-ecr-uri>
```

Step 2: Inspector will Auto-Scan the ECR Image

- Go to **Inspector > Findings**
 - Filter by **ECR** to see image vulnerabilities
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Use Cases

Scenario	Description
CVE Monitoring	Detect Log4j, Heartbleed, etc.
Patch Management	Discover outdated or vulnerable packages
Compliance	Ensure EC2 meets security standards
Container Security	Scan images before deployment to ECS/EKS
Lambda Hardening	Scan Lambda code packages for embedded flaws



Integrations

- **Amazon Security Hub**
 - **EventBridge** (automate actions on findings)
 - **SNS** for alerts
 - **AWS Organizations** for multi-account management
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Pricing

- Pay per scan (based on resource type)
- Free tier available for ECR/Lambda

More details: <https://aws.amazon.com/inspector/pricing>

Interview Questions

1. **What is Amazon Inspector and what resources does it scan?**
 2. **How is Inspector different from GuardDuty?**
 3. **What agent or permissions does Inspector need for EC2?**
 4. **What happens when a new image is pushed to ECR?**
 5. **Can Inspector scan Lambda functions? How?**
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